

Cybersecurity Network Traffic Analysis

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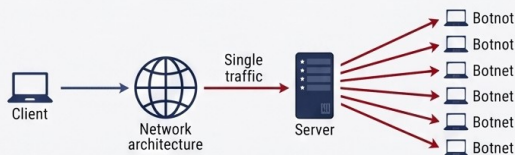
Statistical Analysis of Normal Network Behaviour and DDoS Attack Traffic

Problem Scenario

A company's server traffic has **been** increased from 20,000 requests/hour to 2,000,000 requests/hour during a DDoS attack. Comprehensive analysis of a DDoS attack.

Comprehensive statistical analysis is required, **focusing on a detailed breakdown** and the **identification of DDoS anomalies**.

1. Problem Statement



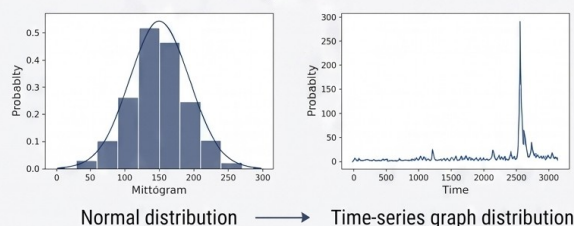
A standard network architecture is **modeled to analyze** a massive traffic stream, identifying specific botnet requests from within massive network traffic data.

2. Data Summary (Descriptive Statistics)

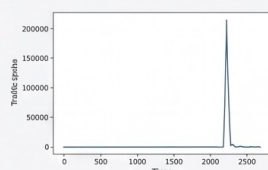
Normal Requests	20,000/hour
Attack Requests	2,000,000/hour
Minimum	18,500
Maximum	2,000,000
Median	20,100
Range	1,981,500
Standard Deviation	Extremely High
Variance	Very Large
Summary	1,970,000
Total Deviation	1,981,500

This summary provides values comparing normal and attack distributions, highlighting stark differences. The attack traffic deviates dramatically from a standard normal distribution.

3. Distribution Analysis



4. Detection of Abnormal Traffic Spikes

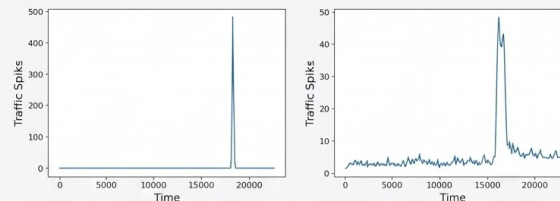


Attack traffic is shifted from a normal ttein a heavily right-skewed down distribution near the exnums extreme peak.

Detection methods rely on identifying sudden abnormal traffic spike; momaus abnormal traffic situation.

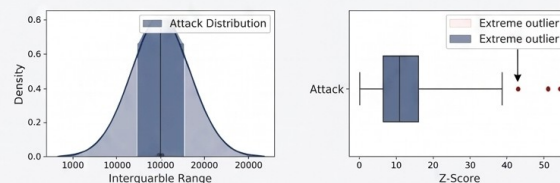
- Z-Score
- Pro extreme Outlier
- Attack Traffic shows extreme observations.

4. Detection of Abnormal Traffic Spikes

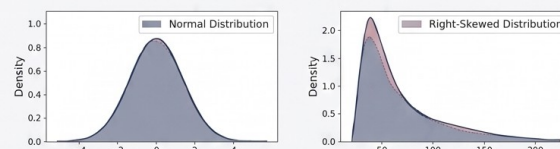


Detection methods utilize interquartile ranges and Z-score analysis to identify abnormal traffic spikes. These sudden spikes in traffic volume within a short period indicate a potential attack. Key methods include:

5. Outlier Detection and Interpretation



6. Skewness and Kurtosis Analysis



Kurtosis and peak analysis helps observe extreme observations and deviation from normalcy.

7. Role of Anomaly Detection in Cybersecurity



Machine learning models are used to **examine** large volumes of network traffic, distinguishing normal application behavior from malicious patterns for automated anomaly detection.

8. Security Measures

- ✓ DDoS Protection
- ✓ Rate Limiting
- ✓ WAF
- ✓ IDS
- ✓ IPS
- ✓ AI-Based Threat Detection
- ✓ Traffic Filtering
- ✓ Load Balancer
- ✓ Real-Time Monitoring
- ✓ Incident Response Plan

9. Key Findings

- ✓ Traffic increase
- ✓ Outliers are distributed
- ✓ Attack successful.

10. Conclusion

This framework concludes with a detailed statistical analysis methodology for DDoS detection. By comparing attack traffic against a normal baseline using statistical measurements, we can ensure the successful **identification and detection summary**.